

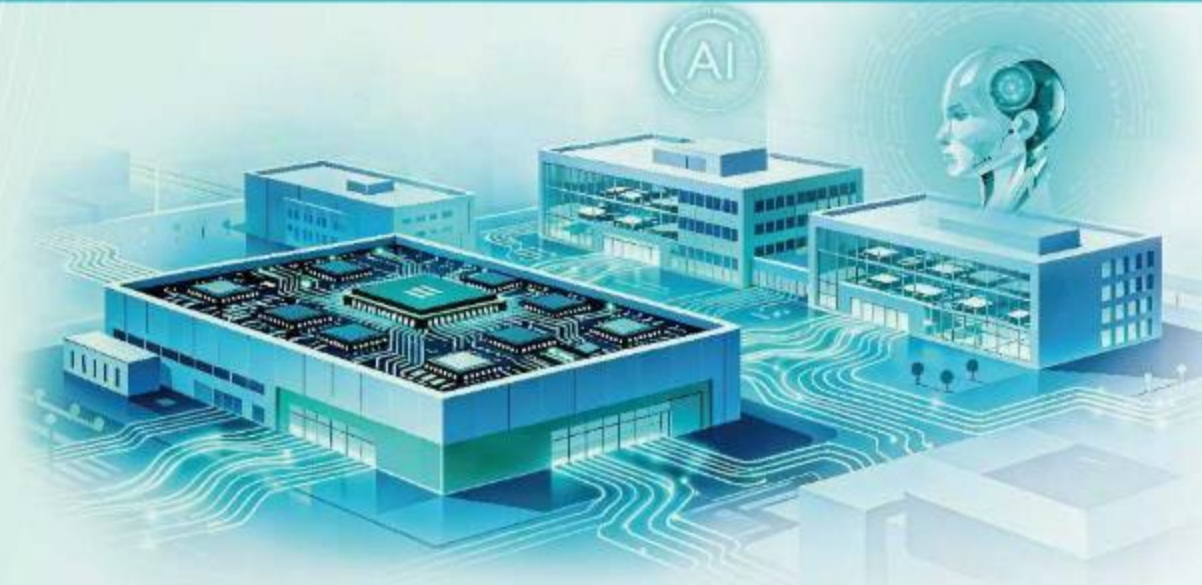


2026 高科技廠房設施國際論壇
High-Tech Facility International Forum 2026

AI-Driven Semiconductor Facility Intelligence:

Leveraging Advanced MPC for Energy Optimization
and Sustainable Transformation

Jack Wang, Rockwell Automation





AI and autonomous systems will change the Facility Control landscape

FROM AUTOMATION

System is programmed to perform tasks in the absence of human intervention

DEFINITION

Programmed

ADAPTATION

Manually re-programmed

TALENT

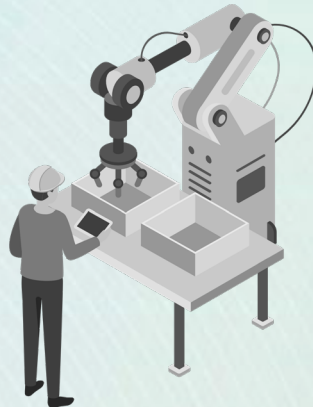
Engineers

KNOWLEDGE ASSETS

Domain knowledge, first principles

BEST APPLICATION

Simple, linear problems with few variables



TO AUTONOMY

System *learns* to perform tasks, and adapts learnings in environments of uncertainty

Programmed + **Learned**

Dynamically adapted
with continuous learning

Engineers +
Data Science Skillsets

Domain Knowledge +
High volume of diverse data

Multi-variable, nonlinear, dynamic problems



 | Traditional Regulatory Control

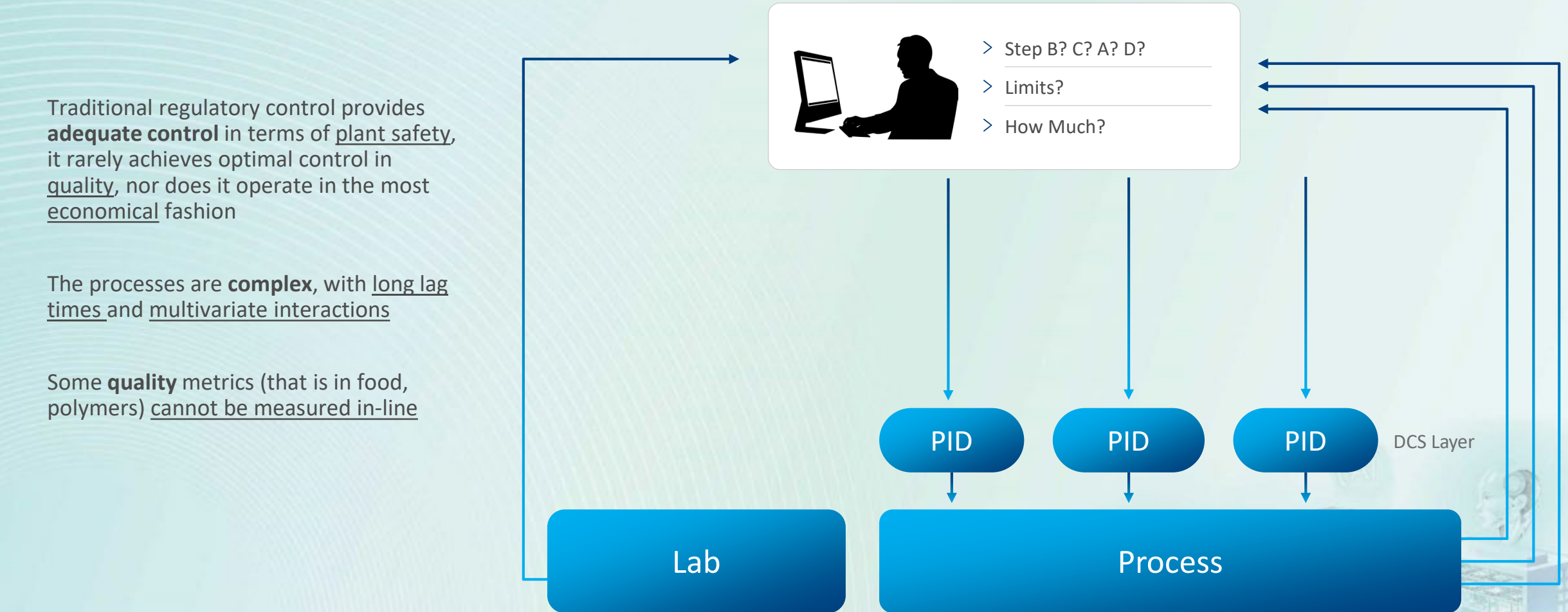
Traditional Regulatory Control

The burden of system optimization falls on the operator

Traditional regulatory control provides **adequate control** in terms of plant safety, it rarely achieves optimal control in quality, nor does it operate in the most economical fashion

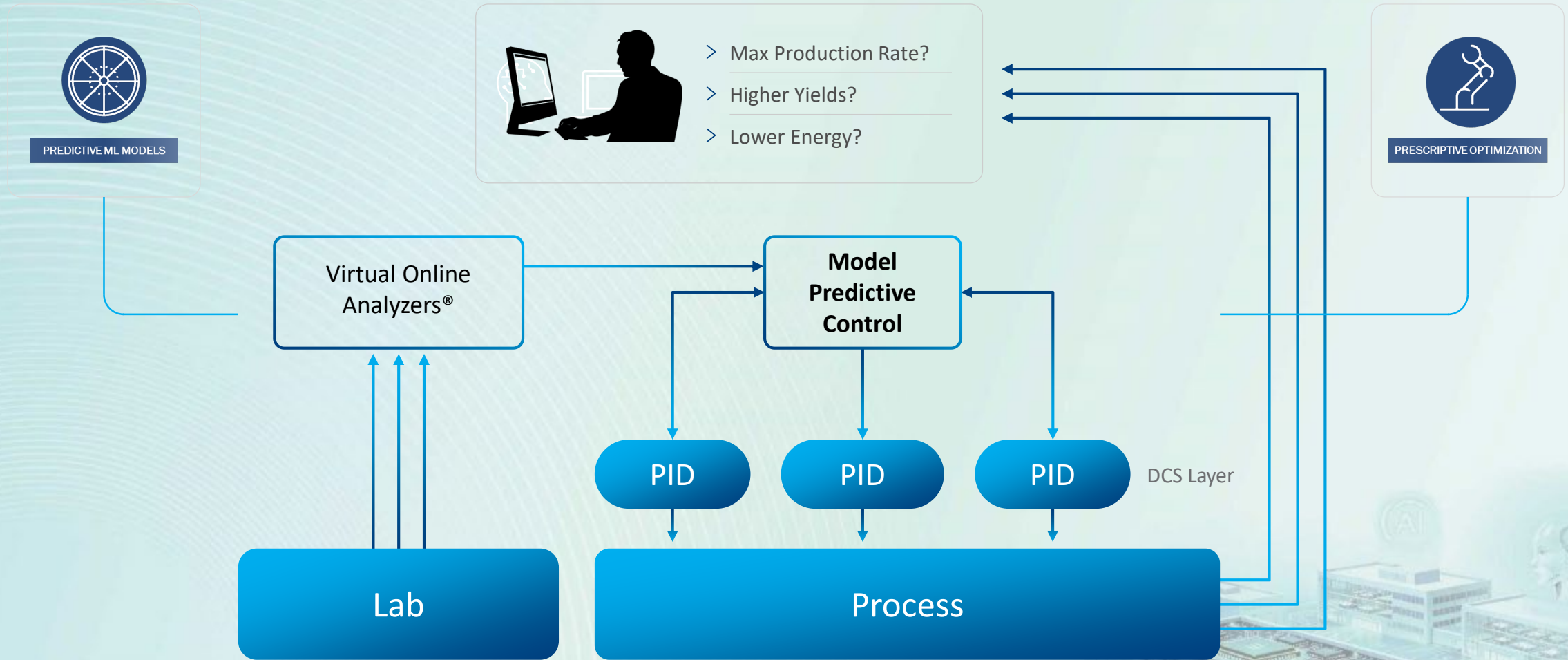
The processes are **complex**, with long lag times and multivariate interactions

Some **quality** metrics (that is in food, polymers) cannot be measured in-line





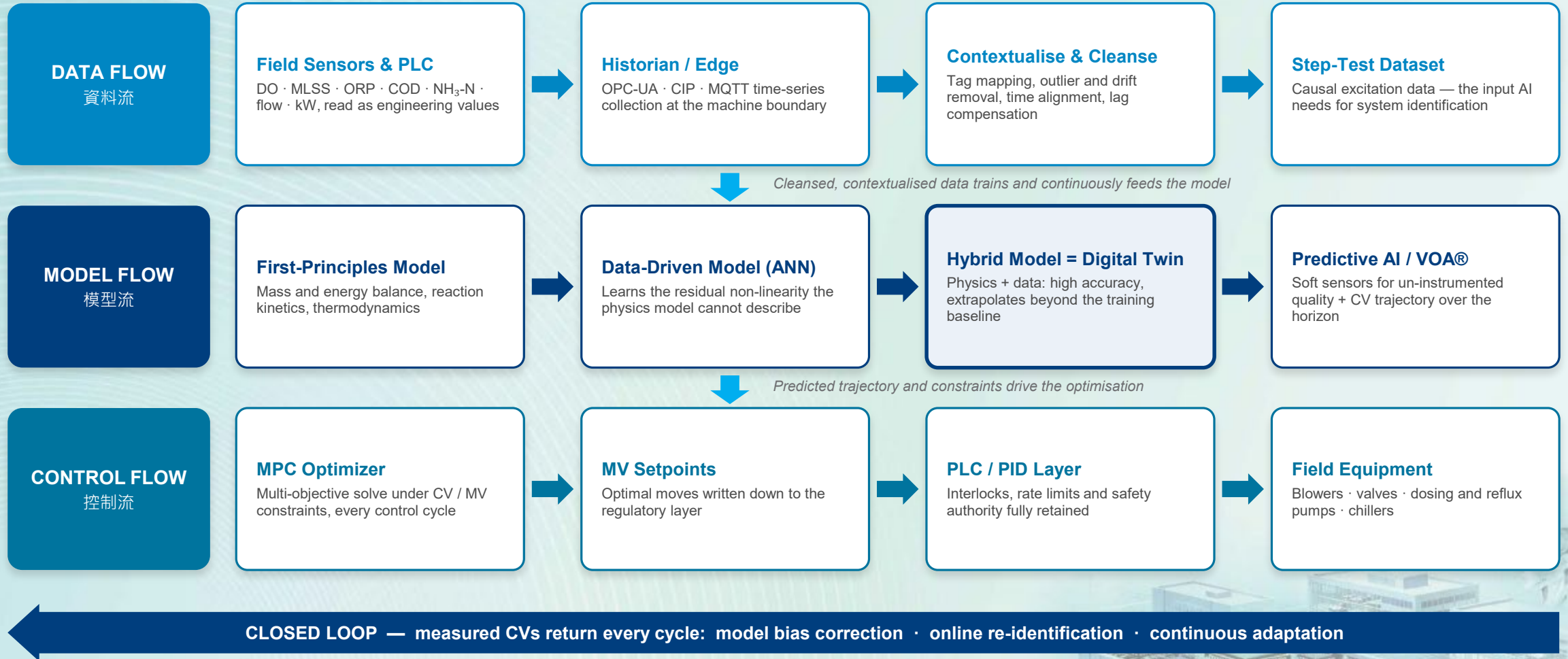
AI Predictive Control Architecture Overview





Overall AI Architecture

How data flow, model flow and control flow connect — Digital Twin | Hybrid Model | Predictive AI





The role of AI in process control & optimization

Hybrid Modeling

01

Physics-Based Models

Physics-based modeling builds on knowledge from scientific principles, for example, thermodynamics, fluid dynamics and/or fluid modeling. Models are less data dependent, extrapolate well and parameters have physical meaning. They are frequently incomplete or limited.

$$\Delta x = A \log(E/E_0)$$

$$CS = 1/2\pi \sqrt{(g/(R-r))}$$



02

Composite Hybrid model

Merging the two principles of physics and data-driven learning combines the best of the two worlds, resulting in higher accuracy, better scalability and cost efficiency.

01

Data-driven Models (incl. ANN)

Based on learning algorithms, which find relationships between sensor data and result variables in a training Data-Set™. Data-driven models require less reasoned knowledge, solve rapidly and match available data well. They are poor at extrapolation and are data-dependent including manufacturing confounded data.

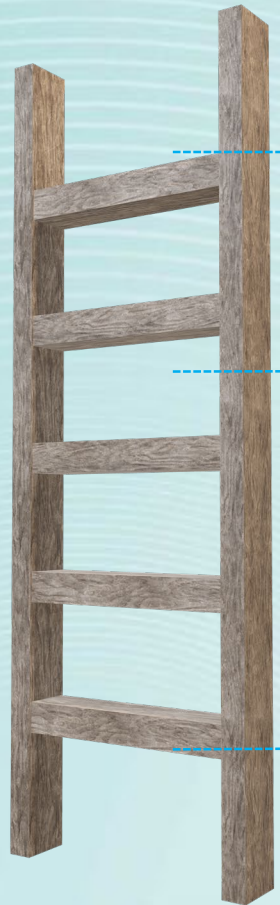


The role of AI in Process Control & Optimization

The importance of causality

“No machine can derive explanations from raw data. It needs a push”

The Book of Why, The New Science of Cause and Effect, J. Pearl & D. Mackenzie



3. COUNTER-FACTUALS

What would have happened with output (**CV**) if I had moved inputs (**MV**) differently?

2. INTERVENTION

How can I reach my target (**CV**) by manipulating the inputs (**MV**)?

1. ASSOCIATION

How are the inputs & outputs correlated?
What will the output be, given inputs? (For example, **VOA®**)

- ▶ Necessary for **prescription: closed-loop control & optimization.**
- ▶ Questions cannot be answered by passively collected (historical) data.
- ▶ Best way to predict the result is to experiment under controlled conditions: **step testing.**
- ▶ Requires a **causal model.** AI uses **control matrix** to encapsulate SME knowledge.

- ▶ This is what most ML/AI tools do
- ▶ Focus on improving mathematical accuracy of prediction. Process gains not relevant.
- ▶ Bad at generalizing. They fit the past well, but **no causality.**
- ▶ Sufficient for Prediction. **Should not be used for control.**

‘The Ladder of Causation’, from ‘The Book of Why’



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The Role of AI in the Control Loop

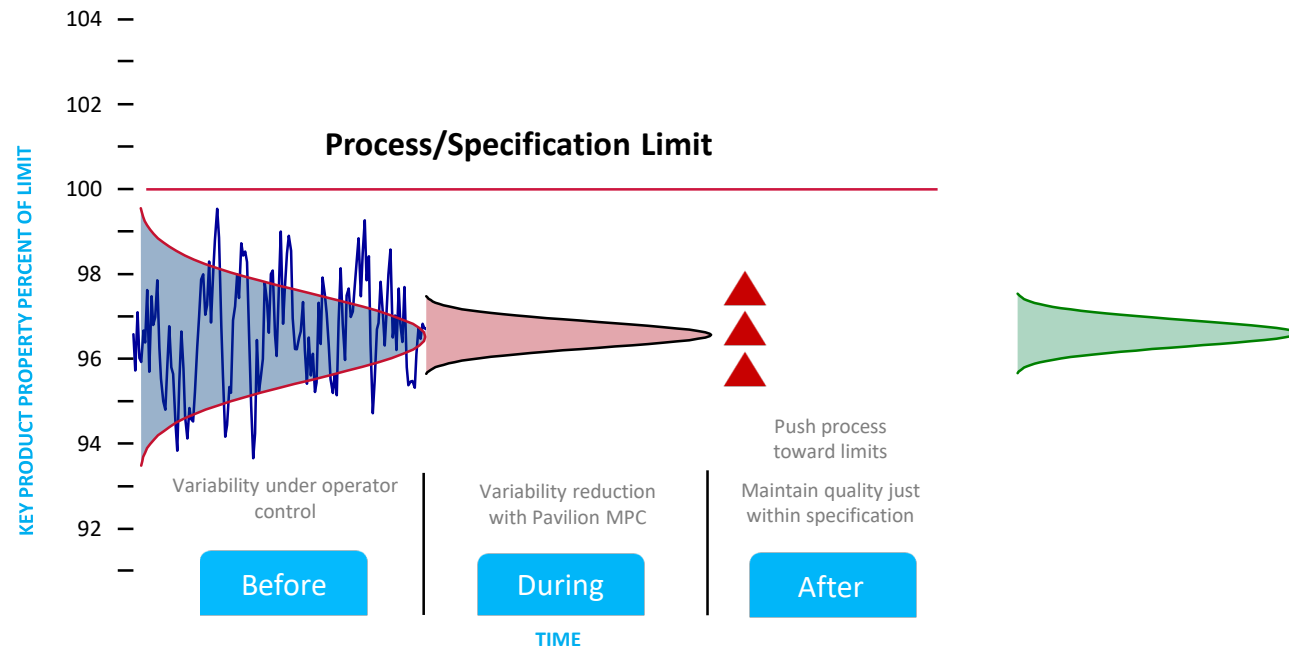
Traditional MPC vs. AI-Driven Predictive Control — where the technical innovation sits

CONTROL-LOOP TASK	TRADITIONAL MPC	AI-DRIVEN PREDICTIVE CONTROL
01 System Identification 系統辨識	Manual step tests at commissioning; a linear FIR / state-space model fitted once, taking weeks of engineering effort.	AI mines historian and step-test data to auto-identify gain, dead-time and non-linear dynamics across the full operating envelope.
02 Hybrid Model Building 混合模型建構	One linear model, valid only in a narrow band around a single operating point.	First-principles structure with an ANN learning the residual — the hybrid model becomes the plant's Digital Twin.
03 Parameter Update 參數更新	Model quality decays with fouling, season and load; recovery needs a manual re-tuning campaign.	Continuous online learning: bias and gains updated every cycle, so the controller self-adapts instead of degrading.
04 Model Prediction 模型預測	Extrapolates measured CVs only; quality that has no on-line instrument is left to the lab.	VOA® soft sensors predict un-instrumented variables (TN, NH ₃ -N, kW/ton) in real time across the full horizon.
05 Optimisation Decision 最佳化決策	Tracks fixed setpoints against a single objective; the economic trade-off is left to the operator.	Multi-objective economic optimisation — energy cost, carbon, quality and equipment life solved simultaneously under constraints.

What does not change: the PLC / PID regulatory layer keeps full authority over interlocks, rate limits and safety. AI writes setpoints into the loop — it never bypasses the protection layer.



How AI Predictive Control Generates Benefits



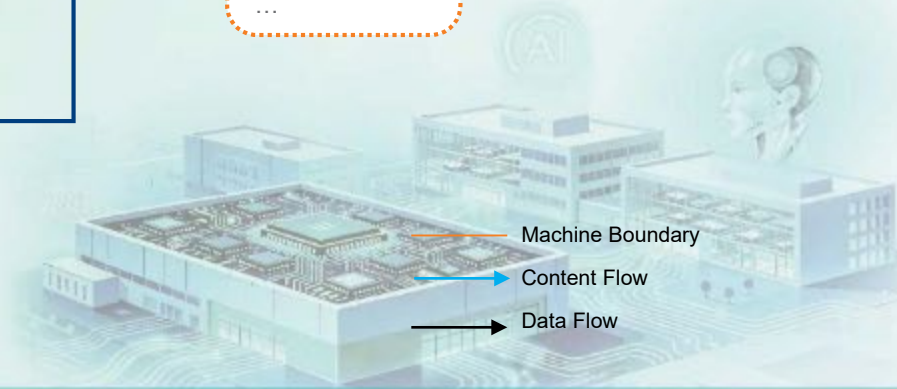
REDUCES variability and manages the process within constraints



ACHIEVES “plant obedience”



ACHIEVES UPLIFT – operate closer to specifications and performance limits while maintaining safety margins





AI Smart Control in a Water Treatment Facility

Case Study

Carbon Source Dosing

Influent C/N Ratio: 7.52 AVG

- **Equipment:**
2 storage tanks, 3 metering pumps (2 duty, 1 standby), and 2 flow meters.
- **Missing Controls:**
No pipeline valves installed; tracking relies strictly on total volume math.
- **Inefficiency:**
Excessive dosing occurs under maximum pump frequency settings.

Sludge & Internal Reflux

High MLSS: 5,000 - 10,000 mg/L

- **Continuous Run:**
4 variable frequency drives running constantly at maximum frequency without feedback.
- **Blind Operations:** Malfunctioning sludge flow meter forces reliance on old historical configurations.
- **No Real-time Feed:** Zero automatic feedback loops or dynamic adjustments.

Aeration & DO Control

Effluent Ammonia: ~0.04 mg/L

- **Empirical Flow:**
Airflow distribution based on static configurations rather than real-time demand.
- **Energy Waste:**
Daily aeration volume averages ~4,000 m³/h, causing severe power drain.
- **Wear & Tear:**
Frequent fan cycling triggers exceptionally high equipment failure rates.

The Vicious Cycle:

Over-Aeration (Empirical High Airflow)

High Dissolved Oxygen (DO) Carryover

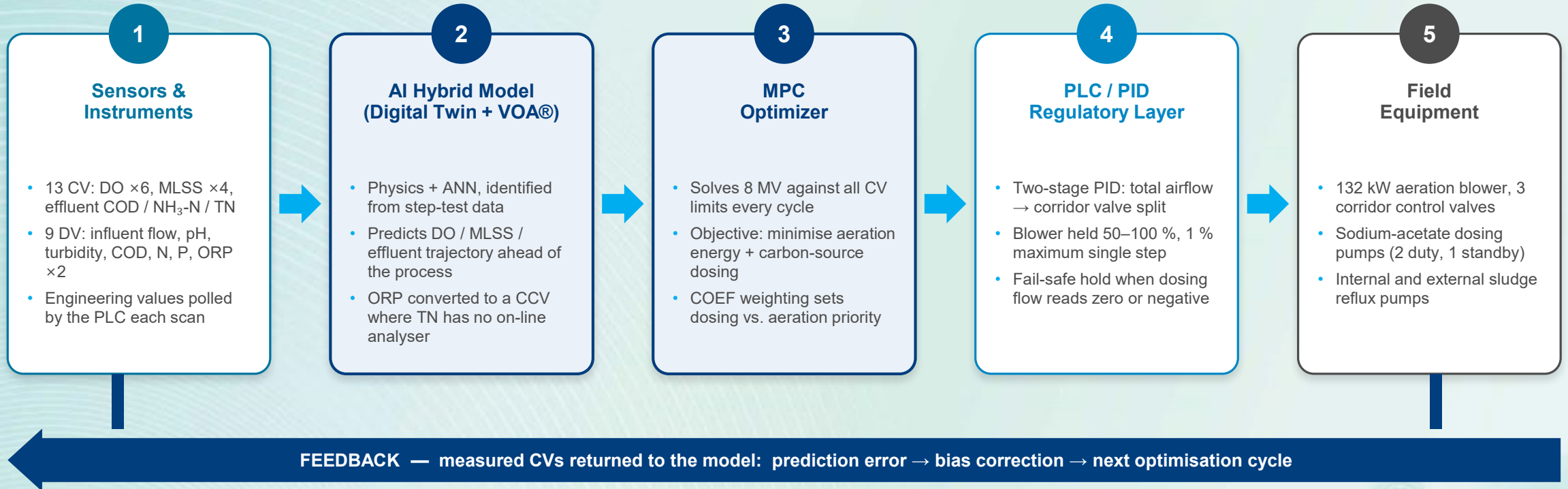
Decreased Denitrification Efficiency

Excessive Carbon Demand (Sodium Acetate Waste)



AI Closed-Loop Control Flow

Sensor → AI Hybrid Model → MPC Optimizer → PLC / PID → Field Equipment → Feedback



Closed-loop cadence: model refresh reduced from 5 min to 10 min per pass without loss of control quality · PID rate limits protect the blower inside the loop · the AI layer can be dropped to manual at any time, leaving regulatory control untouched.



AI Control in a semiconductor water treatment facility

Case Study

Building the model

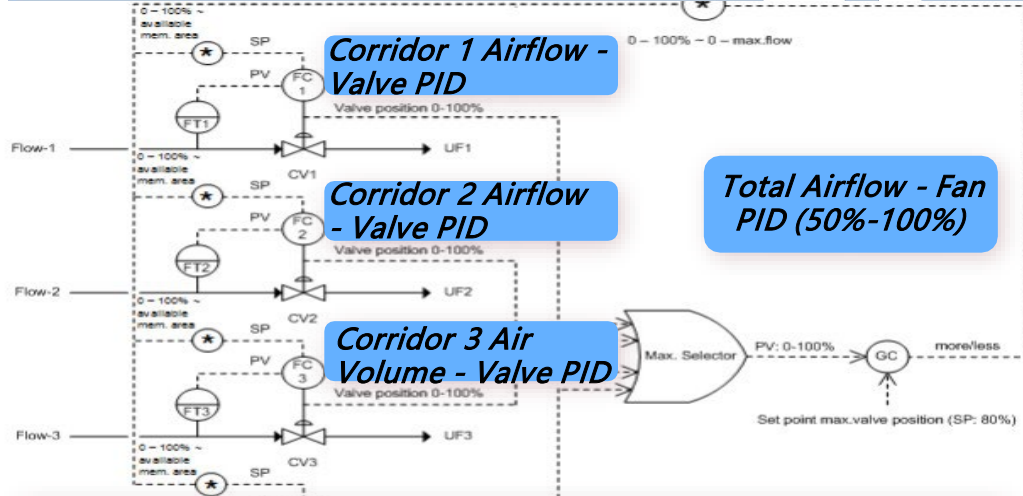
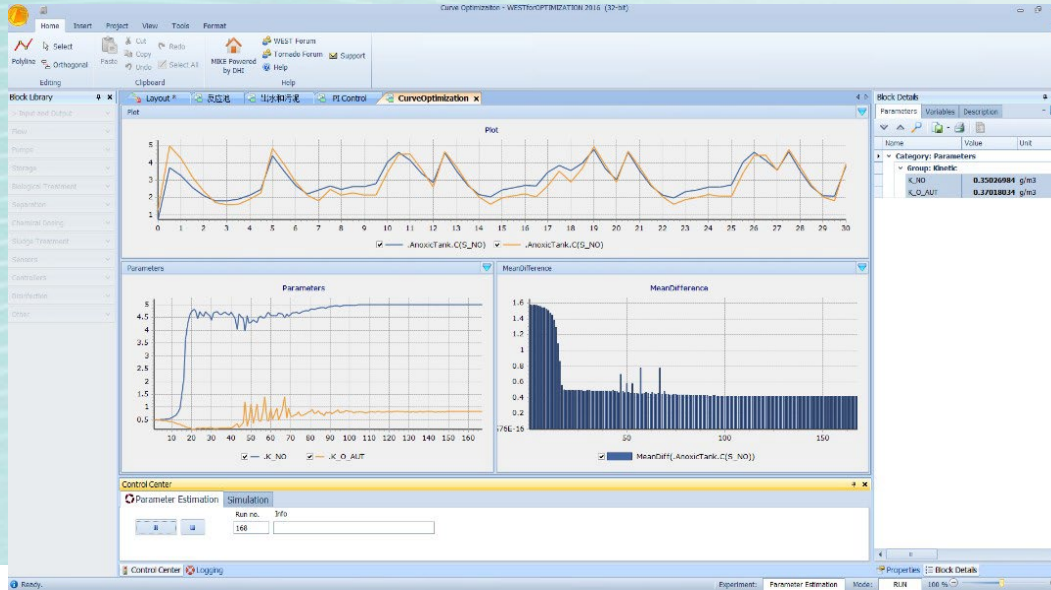
變數	變數描述	PLC可讀取資料來源	AI 控制策略	PLC控制策略	實現功能
CV0	1#溶解氧	生物池1#溶氧儀工程值	參考當前值，降低波動率		減少波動率
CV1	2#溶解氧	生物池2#溶氧儀工程值	參考當前值，降低波動率		減少波動率
CV2	3#溶解氧	生物池3#溶氧儀工程值	控制值比現有控制值降低，降低波動率		減少過量曝氣
CV3	4#溶解氧	生物池4#溶氧儀工程值	參考當前值，降低波動率		減少波動率
CV4	5#溶解氧	生物池5#溶氧儀工程值	參考當前值，降低波動率		減少波動率
CV5	6#溶解氧	生物池6#溶氧儀工程值	控制值比現有控制值降低，降低波動率		減少過量曝氣
CV6	1#池一段MLSS	1#一段缺氧池淤泥濃度工程值	控制值比現有控制值降低，降低波動率		降低回流量
CV7	1#池二段MLSS	1#二段缺氧池淤泥濃度工程值	控制值比現有控制值降低，降低波動率		降低回流量
CV8	2#池一段MLSS	2#一段缺氧池淤泥濃度工程值	控制值比現有控制值降低，降低波動率		降低回流量
CV9	2#池二段MLSS	2#二段缺氧池淤泥濃度工程值	控制值比現有控制值降低，降低波動率		降低回流量
CV10	出水COD	出水COD工程值	參考內控指標，上限卡邊控制		穩定達標
CV11	出水氨氮	出水氨氮工程值	參考內控指標，上限卡邊控制		穩定達標
CV12	出水總氮	出水總氮工程值	參考內控指標，上限卡邊控制		穩定達標
DV0	進水流量	進水流量計工程值	計算對於目標變數的Gain、滯後時間等擾動參數	1#池：2#池按比例拆分	
DV1	進水PH	進水PH計工程值	計算對於目標變數的Gain、滯後時間等擾動參數		
DV2	進水濁度	進水濁度計工程值	計算對於目標變數的Gain、滯後時間等擾動參數		
DV3	進水COD	進水COD測定儀工程值	計算對於目標變數的Gain、滯後時間等擾動參數		
DV4	進水氨氮	進水氨氮測定儀工程值	計算對於目標變數的Gain、滯後時間等擾動參數		
DV5	進水總磷	進水總磷工程值	計算對於目標變數的Gain、滯後時間等擾動參數		
DV6	進水總氮	進水總氮工程值	計算對於目標變數的Gain、滯後時間等擾動參數		
DV7	1#ORP	生物池1#ORP值工程值	計算對於目標變數的Gain、滯後時間等擾動參數		
DV8	2#ORP	生物池2#ORP值工程值	計算對於目標變數的Gain、滯後時間等擾動參數		
MV0	總風量	鼓風機房空氣流量計	AI模型計算輸出	鼓風機PID調節	減少波動，減少過量曝氣
MV1	1#池支管風量配比	1#流量計：2#流量計：3#流量計	AI模型計算輸出	1#調節閥PID+2#調節閥PID+3#調節閥PID	減少波動，減少過量曝氣
MV2	2#池支管風量配比	4#流量計：5#流量計：6#流量計	AI模型計算輸出	4#調節閥PID+5#調節閥PID+6#調節閥PID	減少波動，減少過量曝氣
MV3	1#池乙酸鈉投加泵量	1#乙酸鈉流量計	AI模型計算輸出	乙酸鈉投加泵頻率PID控制	減少過量投加
MV4	2#池乙酸鈉投加泵量	2#乙酸鈉流量計	AI模型計算輸出	乙酸鈉投加泵頻率PID控制	減少過量投加
MV5	1#池內回流流量	1#內回流泵頻率（2#內回流泵頻率）	AI模型計算輸出	內回流泵頻率控制	降低過量回流
MV6	2#池內回流流量	3#內回流泵頻率（4#內回流泵頻率）	AI模型計算輸出	內回流泵頻率控制	降低過量回流
MV7	外回流流量	回流污泥流量計	AI模型計算輸出	污泥回流泵頻率PID控制	降低過量回流



AI Control in a semiconductor water treatment facility

Case Study

Measures to protect the fan in parameter settings to avoid situations such as too low, too fast, or too much adjustment



Airflow control - two-stage PID circuit

MV		
1#_Oxidant_flow		
Name	1#_Oxidant_flow	
EnableIn	True	
ProgAutoReq	False	
ProgManualReq	False	
OperAutoReq	False	
OperManualReq	False	
ManTrajectoryReq	False	
ChangeCoreReq	False	
PredictionReq	True	
ManLimiting	False	
WindupLoIn	False	
WindupHiIn	False	
AOFault	False	
ManValueProg	0	
ManValueOper	0	
SPValueReq	False	
SPTrajectoryReq	False	
SPProg	0	
SPOper	0	
InputFilter		
SPTrajectory		
MaxValue	100	
MinValue	50	
SPHiLimit	100	
SPLoLimit	50	
HiLimit	100	
LoLimit	50	
ROCPosLimit	0.01	
ROCNegLimit	0.01	
Coef	1	
Scale	0.1	
TrajectoryWeight	0	
MoveSupp	1	

Refer to actual operations: the lower limit should not be less than 50% to avoid output being too low

Adjustable 1% of the single stride to avoid excessive adjustment

Lower Scale value reduces fan adjustment operations

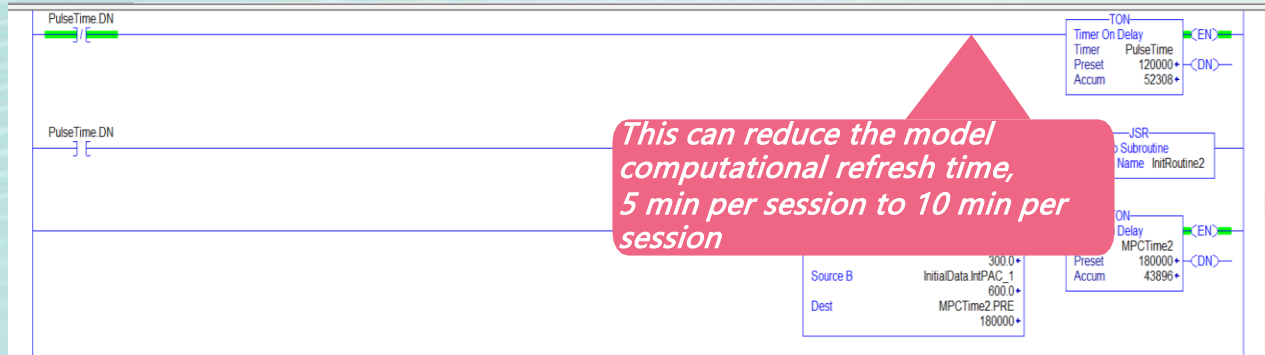


AI Control in a semiconductor water treatment facility

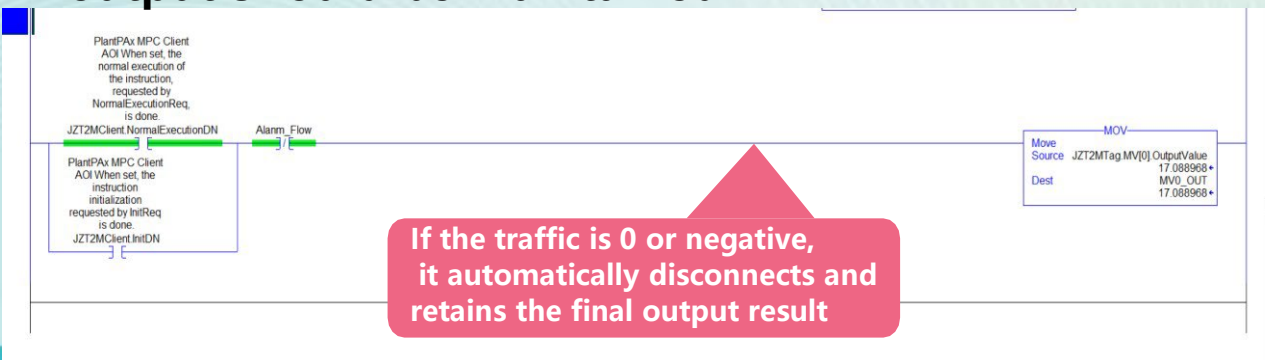
Case Study

Implementation

- ❑ This applies when the target value is already within the normal range, but the dosage has not decreased and remains at the highest level



- ❑ When the dosing pipeline flow is too low, becomes zero, or becomes negative, the current dosing output should be maintained



- ❑ To address the lack of process nitrogen index measurement, ORP can be converted to CCV control

South_ORP	
Name	South_ORP
EnableIn	True
InitReq	False
ChangeCoreReq	False
PredictionReq	True
Measured	450.09
MeasuredFault	False
UseLastOnFault	False
UsePredictionOnFault	False
SPValueReq	True
SPTrajectoryReq	False
ZoneReq	False
ZoneDVRRejectReq	True
SPProg	450
SPOper	450
SPFilter	
SPTrajectory	
MaxValue	1000
MinValue	-1000
SPHiLimit	1000
SPLoLimit	-1000
ZoneHiLimitProg	1000
ZoneLoLimitProg	-1000
ZoneHiLimitOper	1000
ZoneLoLimitOper	-1000
Coef	4
Scale	100

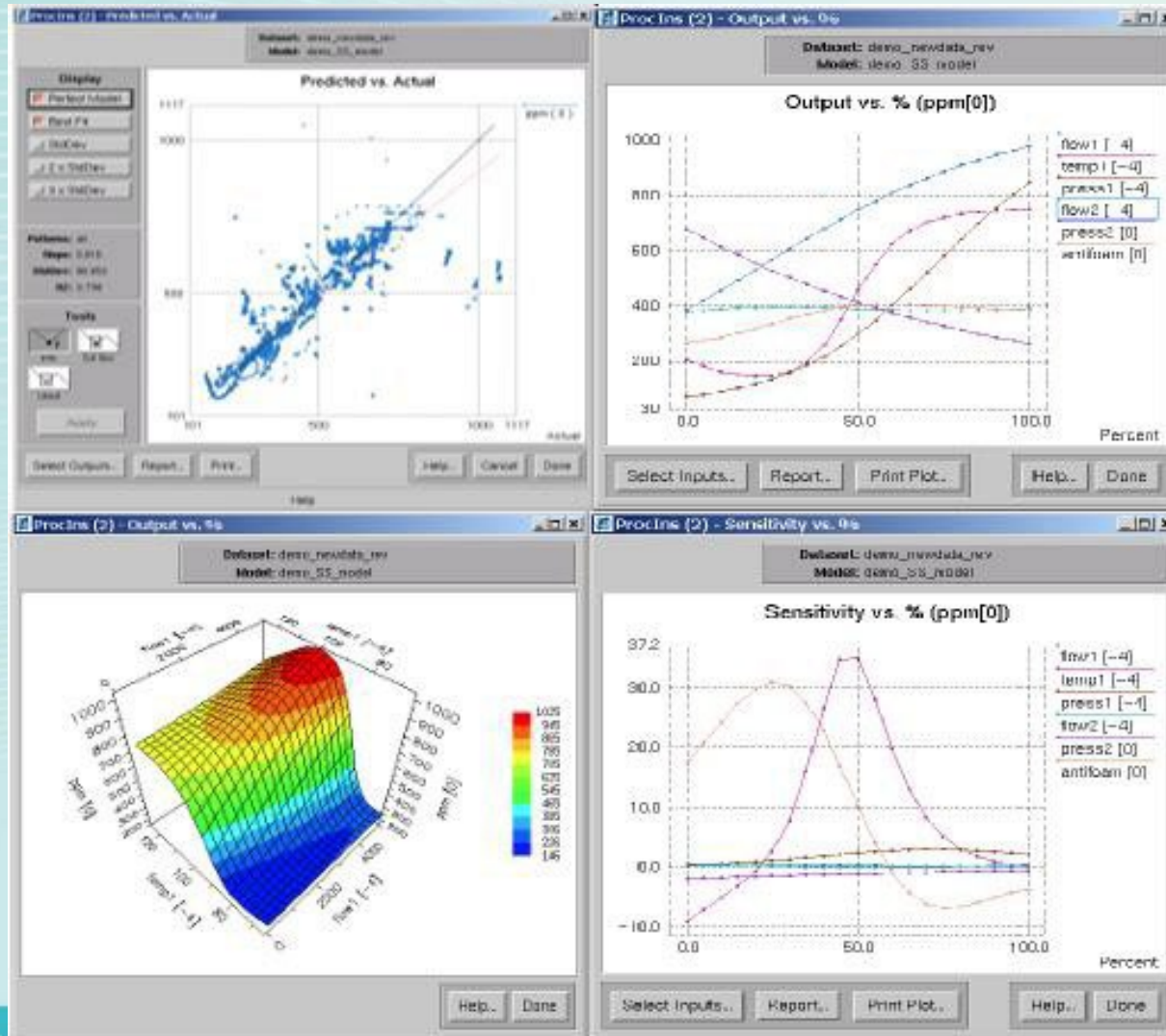
COEF value can be set to adjust the priority of dosing treatment



AI Control in a semiconductor water treatment facility

Case Study

Implementation



- Data communication collection
- Data analysis, data storage, and offline modeling
- Model analysis and simulation debugging
- Online control and optimization
- Online monitoring and KPI performance analysis





AI Control in a semiconductor water treatment facility

Case Study

Performance & Benefit Analysis

Financial Turnaround & ROI

~ 1.5 Years

EXPECTED DIRECT PAYBACK PERIOD

- **NT\$ 4.02 Million+**
in combined direct annual operational savings.
- **Zero Workflow Friction:**
Rapid deployment without process workflow disruption or physical plant modifications.
- **Indirect ROI:**
Additional unquantified savings from reduced labor overhead and optimized process parameters.

Aeration Blower Optimization

NT\$ 403,286 / Yr

DIRECT ENERGY SAVINGS

- **Efficiency Gain:**
Achieved a certified 10% efficiency improvement on a 132 kW blower.
- **Lower Wear & Tear:**
Minimized aggressive fan cycling to mitigate core equipment fatigue.

Chemical Dosing Optimization

NT\$ 3.62 Million / Yr

DIRECT CARBON SOURCE SAVINGS

- **Dosing Precision:**
Attained a complete 10% reduction in annual Sodium Acetate baseline consumption (saving 670.6 Tons/Yr).
- **Process Stability:**
Reduced control variance for Total Nitrogen and Ammonia by over 30%.



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Optimizing Energy Usage of Fab Chiller Systems

Case Study

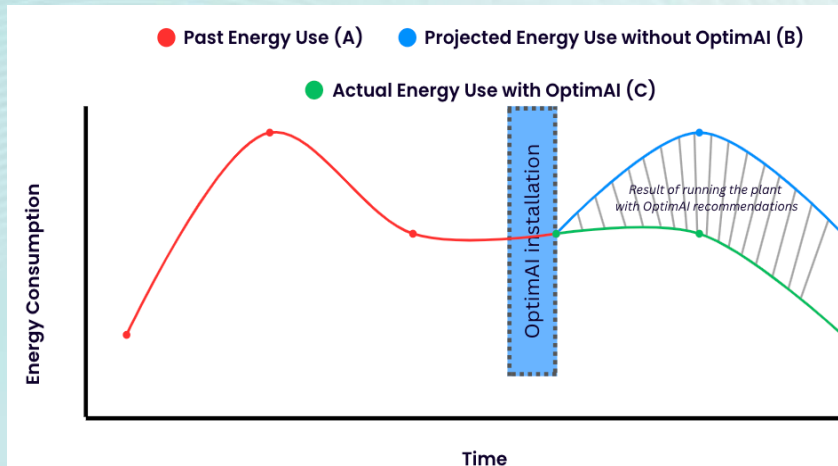
Multinational semiconductor manufacturer.

Business imperatives:

- Sustainability – Reach ambitious multiyear goals to lower GHG emissions, energy and water usage.
- Cost – Reduce annual electricity usage by 10%+.
- Advanced Control – Modernize existing SCADA system to support advanced control and optimization.

Implement artificial intelligence technologies to unlock energy cost savings

- A "digital twin" or data-driven representation of the physical chiller system(s) is built
- Optimization models are trained against the digital twin and real-time process data
- Solution evaluates multiple scenarios, considering operational boundary parameters, identifying the most cost-effective operational plan.
- Models publish recommended control actions to the SCADA system

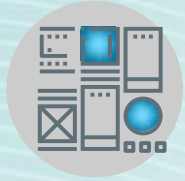


Expected Results

- ✓ Significant improvement in chiller plant efficiency (kW/ton) compared with FY2024 weather-normalized baseline.
- ✓ **\$3-5M** in projected savings, with ROI in less than 2 years.



Key Success Factors for AI Predictive Control Adoption



Data must be extracted from legacy systems



Data must be contextualized & cleansed – AI/ML can help
Data governance and standards are required.



Apply AI/ML at the control level



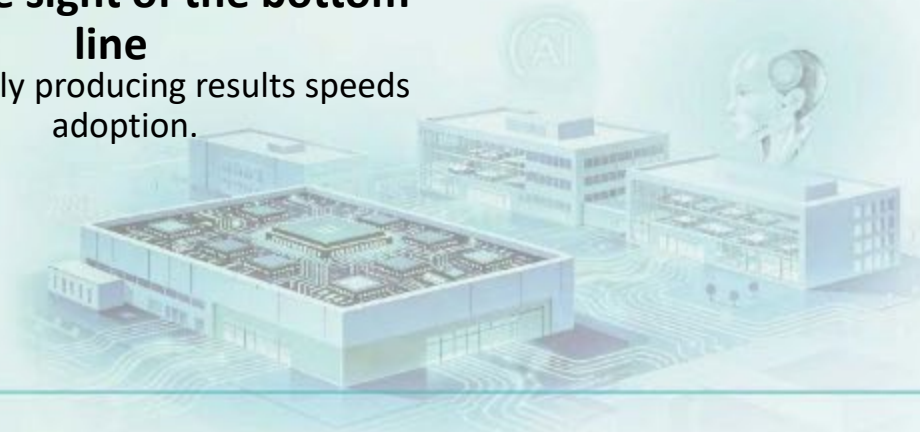
Involve all levels of personnel in problem solving

This helps reduce fear of AI/ML methods.



Don't lose sight of the bottom line

Consistently producing results speeds adoption.



RA | The Smart Path to Predictive AI Control

Narrow Focus, Deep Impact

 DON'T DO THIS	 DO THIS
Foundational capabilities	Full stack of capabilities and value
Generalized across use cases	Narrow, specific scope for priority use case
Scale to add value	Scale to add new applications

 Solutions fail when they don't achieve **immediate business impact**

 Business impact usually requires **comprehensive capabilities**

 Therefore, **foundational approaches tend to fail**

 Alternatively, apply a **comprehensive capability with narrow application focus**

DEPTH OF VALUE
CREATION



**Release 1 –
Narrow Focus**
Value realized for
priority use case

**Prescriptive
modeling,
simulation
and optimization**

**Predictive modeling,
machine learning,
anomaly detection**

**Root cause analysis,
correlation and pattern
detection**

**Data search, reporting and
descriptive analytics abilities**

**Foundational connectivity
architecture and data model**



Release 1 – Foundation Only

BREADTH OF USE CASE APPLICATION



Conclusion: Value for Semiconductor Facility Systems

From control performance to ESG and net-zero contribution

01

Energy Efficiency

能源效率

10 %

certified efficiency gain on a 132 kW blower

- Aeration and dosing follow real-time demand, not a static configuration
- Chiller plant optimised against a weather-normalised FY2024 baseline
- NT\$4.02 M+ direct annual saving; payback ≈ 1.5 years

02

Carbon Reduction

減碳貢獻

≈ 55 tCO₂e

avoided per year on one blower (est.)

- ≈116,000 kWh/yr avoided translates directly into Scope 2 reduction
- 670.6 t/yr sodium-acetate cut lowers the chemical (Scope 3) footprint
- The same models scale to chillers, CDA and exhaust — the largest fab loads

03

Equipment Reliability

設備可靠度

> 30 %

reduction in TN and NH₃-N control variance

- Tighter control means less thermal and mechanical stress on actuators
- Aggressive fan cycling suppressed, reducing blower fatigue
- Rate-limited PID keeps every actuator inside its duty envelope

04

ESG & Net-Zero

淨零目標支援

MRV-ready

auditable KPI baseline for disclosure

- Continuous kW/ton and kWh/m³ KPI baseline, measured not estimated
- Weather-normalised reporting supports SBTi / RE100 disclosure
- Deployed with no process modification — zero workflow friction

AI Predictive Control turns facility systems from a fixed cost centre into a continuously self-optimising asset — delivering measurable energy, carbon and reliability gains that feed directly into the fab's net-zero roadmap.

Source: the semiconductor water-treatment deployment and the fab chiller programme presented in this session. CO₂e figure is an estimate at a 0.474 kg CO₂e/kWh Taiwan grid factor.



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"The technologies are already in place; success depends on effectiveness of integration."

